

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

BIOLOGY HSSC-I (1st Year)

ANNUAL EXAMINATION 2026 (PREDICTED)

Time Allowed: 3:00 Hours

Total Marks: 85

Instructions for Candidates: Answer all questions from Section 'A' on the OMR sheet. Section 'B' and 'C' are to be answered on the provided E-Sheet. Calculators are allowed. Use of lead pencil is strictly prohibited except for diagrams.

SECTION — A (Marks 17)

(Time allowed: 25 Minutes)

Q. 1 Choose the correct answer by filling the relevant bubble on the OMR sheet.

1. A cell with 0.5M glucose concentration is placed in a 0.2M glucose solution. The cell will become:

- A) Flaccid B) Turgid C) Plasmolyzed D) Lysed

2. Which of the following organelles is directly involved in the synthesis of lipids and detoxification of drugs?

- A) RER B) SER C) Golgi Complex D) Lysosomes

3. In competitive inhibition, the inhibitor binds to the:

- A) Allosteric site B) Active site C) Enzyme-product complex D) Co-factor

4. The secondary structure of protein (α -helix) is maintained by:

- A) Peptide bonds B) Ionic bonds C) Hydrogen bonds D) Disulphide bridges

5. Which of the following is a non-reducing sugar?

- A) Glucose B) Fructose C) Maltose D) Sucrose

6. The final electron acceptor in the non-cyclic electron flow (Z-scheme) is:

- A) P700 B) NADP⁺ C) Oxygen D) Water

7. A bacteriophage consists of:

- A) DNA and Capsid B) RNA and Envelope C) Protein only D) DNA and Ribosomes

8. The cell wall of Gram-positive bacteria contains a high percentage of:

- A) Lipids B) Peptidoglycan C) Cellulose D) Chitin

9. Which phase of bacterial growth shows the maximum rate of reproduction?

- A) Lag phase B) Log phase C) Stationary phase D) Decline phase

10. Protists that have silica in their cell walls and are major primary producers in oceans are:

- A) Diatoms B) Dinoflagellates C) Euglenoids D) Slime molds

11. Which pigment is essential for the initiation of photosynthesis?

- A) Chlorophyll b B) Xanthophyll C) Chlorophyll a D) Carotene

12. The source of oxygen evolved during photosynthesis is:

- A) CO₂ B) H₂O C) Glucose D) ATP

13. In the life cycle of a fern, the dominant generation is:

- A) Gametophyte B) Sporophyte C) Prothallus D) Antheridium

14. Which of the following is an example of an acoelomate animal?

- A) Hydra B) Planaria C) Roundworm D) Earthworm

15. The enzyme 'Pepsin' acts on:

- A) Carbohydrates B) Lipids C) Proteins D) Nucleic acids

16. 'Double circulation' is first observed in:

- A) Fishes B) Amphibians C) Reptiles D) Birds

17. T-cells are matured in:

- A) Bone marrow B) Spleen C) Thymus D) Lymph nodes

SECTION — B (Marks 42)

(Time allowed: 2 Hours 35 Minutes)

Q. 2 Attempt any FOURTEEN parts from the following. All parts carry equal marks. (14 x 3 = 42)

- i. Differentiate between Lysogenic and Lytic cycles of viruses.
- ii. Sketch the Fluid Mosaic Model of the cell membrane and label three parts.
- iii. Why is ATP called the "Energy Currency" of the cell?
- iv. Compare the cell wall of Gram-positive and Gram-negative bacteria.
- v. Describe the role of 'induced fit model' in enzyme action.
- vi. Mention any three characteristics of Phylum Arthropoda.
- vii. Define 'Emulsion' and give its biological importance.
- viii. Differentiate between Photophosphorylation and Oxidative Phosphorylation.
- ix. How does the 'pressure flow mechanism' explain the translocation of food?
- x. What is 'Photorespiration'? Why is it considered a wasteful process?
- xi. Briefly explain the 'Immunity' provided by B-lymphocytes.
- xii. Describe the structure and function of 'Stomata'.
- xiii. List any three major differences between Monocots and Dicots.
- xiv. Define 'Heat capacity' of water and its importance for living organisms.
- xv. What are 'Prions'? How do they differ from Viroids?
- xvi. Sketch and label the structure of a Mitochondrion.
- xvii. Define 'Feedback Inhibition' with one example.
- xviii. What is the role of 'Bile' in human digestion?
- xix. Differentiate between 'Systemic' and 'Pulmonary' circulation.
- xx. Define 'Endocytosis' and differentiate between Phagocytosis and Pinocytosis.

SECTION — C (Marks 26)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 13 = 26)

Q. 3

- a) Explain the Watson-Crick model of DNA structure in detail with a labeled diagram. (7)
- b) Describe the various steps of 'Glycolysis' (Preparatory and Pay-off phases). (6)

OR

- a) Discuss the light-independent reactions (Calvin Cycle) of photosynthesis. (7)
- b) Explain the 'Mechanism of Enzyme Action' using the lock and key model. (6)

Q. 4

- a) Describe the structure and life cycle of 'Plasmodium' causing Malaria in humans. (7)
- b) Explain the 'Cohesion-Tension Theory' for the ascent of sap in plants. (6)

OR

- a) Discuss the 'Phases of Growth' in bacteria with the help of a growth curve. (7)

b) Describe the structure and functions of 'Nucleus' in a eukaryotic cell. (6)

Q. 5

a) Explain the digestion of food in the 'Human Small Intestine'. (7)

b) Discuss the classification of proteins based on their functional roles. (6)

OR

a) Describe the structure and working of 'Human Heart' with a labeled diagram. (7)

b) Discuss the characteristics of 'Kingdom Fungi' and their economic importance. (6)

SOLUTIONS & ANSWER KEY

Section A - Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
B	B	B	C	D	B	A	B	B	A	C	B	B	B	C	B	C

Section B - Model Answers (Key Points)

i. Lysogenic vs Lytic: Lytic cycle leads to death/lysis of host cell; Lysogenic involves integration of viral DNA into host genome (prophage) without immediate lysis.

ii. Fluid Mosaic Model: Phospholipid bilayer (fluid), Proteins (mosaic: integral/peripheral), Carbohydrates (glycolipids/glycoproteins), Cholesterol.

iii. ATP as Energy Currency: High-energy phosphate bonds; easily broken to release 7.3 kcal/mol; links endergonic and exergonic reactions.

iv. Gram +ve vs -ve: +ve has thick peptidoglycan, teichoic acid, stains purple; -ve has thin peptidoglycan, outer lipopolysaccharide layer, stains pink/red.

v. Induced Fit Model: Active site is not rigid; it changes shape slightly to fit the substrate more tightly for better catalysis.

Section C - Key Explanations

DNA Structure: Double helix, anti-parallel strands, sugar-phosphate backbone, nitrogenous bases (A-T, G-C) held by H-bonds.

Glycolysis: Glucose (6C) broken into 2 Pyruvate (3C). Prep phase uses 2 ATP; Pay-off phase produces 4 ATP (net 2 ATP) and 2 NADH.

Cohesion-Tension: Cohesion (water-water), Adhesion (water-xylem), Transpiration pull creating negative pressure to lift water column.